

OFFSITE — Drone-Based Construction Visibility Platform

An end-to-end UAV-LiDAR monitoring system that transforms aerial site data into real-time 3D intelligence — eliminating the schedule delays and cost overruns that plague 98% of megaprojects.

Total Budget	LiDAR Accuracy	Coverage / Flight	Data Preview
\$3.94M	± 2 cm	2.5 km²	< 10 min
24-month program	RTK/GNSS precision	55-min flight time	Near-complete in 2h

INDUSTRY PROBLEM 98% of megaprojects face cost or schedule overruns 53% of projects finish late — avg delay: 20 months 66% of projects exceed their original budget	OFFSITE SOLUTION Real-time LiDAR aerial data to 3D dashboard in minutes Automated EC2 + Pix4D pipeline — no manual wrangling BIM/CAD-compatible with Procore sync and RBAC
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Project Timeline

Five-phase delivery over 24 months · Agile sprints with bi-weekly retrospectives

Phase	Name	Duration	Key Focus	Status
01	Initiation	M1–2	Feasibility · Stakeholders · Charter · Risk Baseline	Complete
02	Planning	M2–4	Architecture · Tech Stack · Scheduling · Compliance	Complete
03	Execution	M4–18	UI/UX · Backend · Frontend · LiDAR Integration · QA	Active
04	Monitoring	M18–21	CI/CD · Azure Deploy · App Store · User Training	Upcoming
05	Closure	M21–24	Security Audit · Handover · Post-Launch · KT Sessions	Upcoming

Key Milestones

M1	Project Charter signed · Feasibility study complete · Stakeholder map finalized
M4	System architecture approved · Tech stack locked · AWS infrastructure provisioned
M8	Backend APIs complete · LiDAR ingestion pipeline operational
M12	Frontend dashboard live · 3D point cloud rendering validated
M16	Full system integration · UAT with construction firm stakeholders
M18	Security audit cleared · CI/CD pipeline automated · Deployment ready
M21	Go-live · User training complete · Support desk operational
M24	Post-launch review · Final report · Knowledge transfer archived

Financial Plan

Total program investment: \$3,940,393 over 24 months

Category	Amount	% of Total	Description
Labor	\$2,974,400	75.5%	13 roles · Full-stack, DevOps, Drone, QA, PM, Security
Contingency	\$624,023	15.8%	15% buffer · Scope changes, hardware failures, delays
Hardware	\$251,698	6.4%	6x DJI drones, 6x LiDAR sensors, RTK, laptops, tablets
Software	\$90,272	2.3%	AWS infra, Pix4D, GitHub Enterprise, Jira, Figma, Sentry
TOTAL	\$3,940,393	100%	24-month end-to-end platform development

Labor Cost Breakdown — \$2,974,400 · 4,160 hours per role

Role	Rate/hr	Total	Role	Rate/hr	Total
Sr. Full Stack Dev x2	\$75	\$624,000	QA Automation Engineer	\$45	\$187,200
Data Engineer	\$60	\$249,600	Project Manager	\$45	\$187,200
Security Architect	\$60	\$249,600	Frontend Developer	\$45	\$187,200
DevOps Engineer	\$60	\$249,600	UI/UX Designer	\$40	\$166,400
Mid-Level Backend Dev	\$50	\$208,000	Drone Pilot	\$40	\$166,400
Scrum Master	\$50	\$208,000	Technical Writer	\$30	\$124,800

Risk Assessment & Mitigation

14 identified risks across 6 categories · Bi-weekly Jira sprint reviews

Risk	Category	Likelihood	Impact	Rank	Mitigation Strategy
Hardware Failure	Operational	Medium	High	HIGH	Redundant drones + preventive maintenance
Integration Delays	Technical	High	Medium	HIGH	Phased modules + early API testing
Budget Overruns	Financial	Medium	High	HIGH	15% contingency buffer + monthly review
Unauthorized Access	Security	Medium	High	HIGH	MFA + role-based access controls
Network Instability	Operational	High	Medium	HIGH	Offline fallback + pre-flight signal checks
Regulatory Restrictions	Compliance	Medium	High	HIGH	Early FAA permits + GIS no-fly verification
Resource Burnout	Workforce	High	Medium	HIGH	Jira workload tracking + task rotation
Adoption Resistance	Operational	Medium	Medium	MEDIUM	Training sessions + pilot ROI showcase
Skill Mismatch	Workforce	Medium	Medium	MEDIUM	Mentors + drone/cloud training
Encryption Failure	Security	Low	High	LOW	AES-256 + regular security audits
Flight Accidents	Operational	Low	High	LOW	Pre-flight checks + pilot certification
Data Governance	Compliance	Low	High	LOW	Data anonymization + compliance reviews

Resource Allocation & Technology Stack

Team Utilization by Phase (%)

Phase	Project Mgmt	Backend Dev	Frontend Dev	DevOps/Cloud	Drone Ops	QA/Testing
Initiation	80%	10%	10%	20%	30%	0%
Planning	70%	30%	20%	40%	20%	10%
Execution	50%	90%	90%	80%	70%	60%
Monitoring	60%	50%	40%	90%	50%	90%
Closure	80%	20%	20%	30%	20%	40%

Full Technology Stack

Component	Technology / Tool	Component	Technology / Tool
Drone Hardware	DJI Matrice 350 RTK + Zenmuse L2 LiDAR	Frontend Web	React, Three.js, Potree, CesiumJS
LiDAR Processing	Pix4Dmapper, Pix4Dsurvey, DJI Terra	Mobile	Flutter (iOS + Android)
Geospatial Tools	GDAL / PDAL / PostGIS	Cloud Infra	AWS S3, EC2, RDS, CloudFront, VPC
Backend	Node.js, Express, AWS Lambda, EC2	Security	AES-256, OAuth 2.0, JWT, RBAC, MFA
Database	PostgreSQL + PostGIS on AWS RDS	CI/CD	GitHub Actions + Jenkins
Processing Pipeline	AWS Lambda + EC2 + Pix4D automated	Monitoring	Sentry, AWS CloudWatch
Collaboration	Jira, Figma, Slack, GitHub Enterprise	Compliance	FAA Part 107, GDPR, CCPA, NEPA

